NU-SentEval at DialectSentEval Shared Task Subtask-2: Fine-Tuning AraT5 for Arabic Dialect Sentiment Swap

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Abstract

Sentiment style transfer aims to reverse a sentence's sentiment polarity while preserving its content. This task remains underexplored for dialectal Arabic due to non-standardized orthography, code-switching, and dialect-specific variation. We present our system for **DialectSentEval 2026 Shared Task Subtask 2: Arabic Dialect Sentiment Swap**, in which we fine-tune **AraT5v2-base-1024** as an end-to-end sequence-to-sequence model mapping a `(source_polarity, source sentence)` pair to a sentiment-inverted target sentence, using 4-beam search. Our system achieves **55.60 BLEU, 72.27 chrF, 77.90% sentiment preservation** on the Development phase, and **54.63 BLEU, 72.61 chrF, 68.98% sentiment preservation** on the Final Evaluation phase. Manual analysis shows strong performance on short, simple sentences, with occasional failures where secondary cues (e.g., emojis) remain unchanged after the polarity-bearing word is inverted. The drop in sentiment preservation between phases is likely partly due to missing gold `source_polarity` labels in the Final Evaluation set, requiring automatic prediction before generation.

1 Introduction

Sentiment style transfer rewrites a sentence to reverse its sentiment while preserving its content. It remains underexplored for dialectal Arabic, where non-standardized orthography and dialectal variation add extra challenges. The DialectSentEval 2026 Shared Task addresses this through Subtask 2, *Arabic Dialect Sentiment Swap*, which requires rewriting a dialectal sentence to express the opposite polarity while preserving its meaning. Performance is measured by content preservation (BLEU, chrF) and sentiment reversal (via an external classifier).

We present the NU-SentEval team's submission to Subtask 2. We fine-tune **AraT5v2-base-1024** as an end-to-end sequence-to-sequence model that generates a sentiment-inverted sentence from a `(source_polarity, source sentence)` pair, decoded with 4-beam search. We evaluate the system on both shared task phases, reporting BLEU, chrF, and sentiment-preservation scores, along with a manual error analysis. The model performs well on short sentences but sometimes leaves secondary cues, such as emojis, unchanged. The Final Evaluation phase also lacked gold `source_polarity` labels, requiring automatic prediction that may introduce errors.

2 Background

DialectSentEval 2026 Subtask 2 (Ezzini et al., 2026a) provides dialectal Arabic sentence pairs annotated with source sentiment polarity (positive/negative) and, in the Development phase, a gold sentiment-inverted target. Data comes from the MA'AKS dataset (Mughaus et al., 2026), part of the broader AraSentEval framework (Ezzini et al., 2026b) for Arabic sentiment generation tasks. Systems must generate a sentence expressing the opposite polarity while preserving topic and content. Evaluation uses two axes: *content preservation* (BLEU, chrF against the source) and *sentiment preservation* (% of outputs classified as successfully flipped by an external classifier). Submissions require a `predictions.zip` containing `predictions.xlsx` with `id` and `style` columns. The Final Evaluation test set omits the gold target and polarity label, requiring participants to predict polarity automatically before generation.

3 System Overview

We cast Subtask-2 (Sentiment Swap) as a conditional sequence-to-sequence generation problem: given a dialectal Arabic sentence and its current sentiment polarity, the model must generate a new sentence that expresses the *opposite* polarity while

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preserving as much of the original content as possible.

3.1 Base Model

We fine-tune **AraT5v2-base-1024** (Elmadany et al., 2023), an encoder–decoder Transformer following the T5 architecture and pre-trained on large-scale Arabic data (Modern Standard Arabic and dialectal varieties) with an extended input sequence length of 1,024 tokens.https://huggingface.co/UBC-NLP/AraT5v2-base-1024 We use the model's slow, SentencePiece-based tokenizer, which we found necessary for compatibility with the pinned `transformers` version (the fast Rust tokenizer failed to load AraT5's legacy vocabulary).

3.2 Input/Output Formulation

We convert every training pair (`source`, `source_polarity`, `target`) into a single prompt-conditioned input string. The source polarity is injected as an explicit Arabic tag so the model always knows *which direction* to invert:

اعكس المشاعر (<polarity>): <source>

where <polarity> is either إيجابي or سلبي, depending on `source_polarity`. The decoder target is simply the gold `target` sentence, i.e. the same sentence rewritten with inverted sentiment. This prompt-based conditioning lets a single model handle both directions of the swap (positive→negative and negative→positive) without separate models or output heads.

3.3 Decoding

At inference time we generate with standard beam search (4 beams, no sampling), capping generation at the same maximum length used for training targets. No re-ranking or post-editing is applied to the raw model output.

4 Experimental Setup

4.1 Data

We use the official DialectSentEval 2026 Subtask-2 (Sentiment Swap) splits. Table 1 reports the size of each split used in our pipeline.

Split	# examples
Train (official)	6,752
– used for updates	6,077
– internal dev (10%)	675
Development (official val)	964
Final Evaluation (test)	1,931

Table 1: Data splits used for training and evaluation.

The official training set is moderately imbalanced toward negative-source examples (3,731 negative vs. 3,021 positive); we did not apply reweighting or oversampling in our submitted system (see Limitations). We hold out a random 10% of the official training set (675 examples, fixed seed) purely to monitor chrF during training for checkpoint selection; the official Development file is reserved for generating the Development-phase submission, and the official test file is used only once for the Final Evaluation submission.

An important asymmetry between the two phases is that the released Development-phase file includes the gold `source_polarity` label for every example, whereas the real Final Evaluation test file provides only `id` and `source`. For the Final Evaluation submission we therefore predict `source_polarity` ourselves with the same CAMeL sentiment classifier used for scoring (Section 4, below) before constructing the model input, since a positive/negative tag is required to tell the model which direction to invert.

4.2 Preprocessing

Preprocessing is minimal by design, consistent with a pre-trained text-to-text model: no manual normalization, diacritic stripping, or segmentation is applied beyond the tokenizer's own SentencePiece tokenization, so that dialectal spelling, code-switching, and emojis in the input are preserved as signal for the model. Inputs and targets are truncated to a maximum of 96 sub-word tokens each, which comfortably covers the length of the (mostly single- to few-sentence) social-media-style data in this task.

4.3 Training Configuration

The model is fine-tuned with the Hugging Face `Seq2SeqTrainer`. Table 2 lists the hyperparameters used for our submitted system.

We select the checkpoint with the highest chrF (Popović, 2015) on the internal 675-example devel-

Hyperparameter	Value
Epochs	8
Train batch size	16
Eval batch size	32
Learning rate	3e-4
Weight decay	0.01
Warmup ratio	0.06
Max input length	96 tokens
Max target length	96 tokens
Beam size (decoding)	4
Precision	fp16
Seed	42

Table 2: Fine-tuning hyperparameters.

opment split (evaluated once per epoch) as our final model, rather than the last training epoch, using `predict_with_generate` with the same 4-beam decoding used at test time.

4.4 Hardware and Software

Training was run on a single Kaggle GPU accelerator (T4/P100) and took approximately 40 minutes end-to-end for 8 epochs. We use `transformers` 4.46.3, `tokenizers` 0.20.3, `datasets` 3.1.0, and `accelerate` 1.1.1; these specific, pinned versions were required because more recent `transformers`/`tokenizers` releases fail to load AraT5's legacy SentencePiece tokenizer.

4.5 Evaluation Metrics

Following the shared task specification, we report: (i) **BLEU** (Papineni et al., 2002) and **chrF** (Popović, 2015) computed between the *input* (`source`) and the *generated output* (`style`) as a measure of content preservation, and (ii) **Sentiment Style Accuracy**, the percentage of generated outputs whose predicted polarity — according to CAMELBERT-DA (Inoue et al., 2021) sentiment classifier — matches the expected opposite of the source polarity. These are computed identically for the Development and Final Evaluation phases, with the only difference being that `source_polarity` is gold-provided in the former and classifier-predicted in the latter (Section 4).

5 Results

Table 1 reports the official platform results for our fine-tuned AraT5v2-base-1024 system across both evaluation phases of the shared task.

We discuss a likely cause for the drop in Sentiment Preservation between the two phases in the Discussion and Error Analysis and Limitations sections below.

Phase	BLEU	chrF	Sentiment Preservation
Development (964)	55.60	72.27	77.90%
Final Evaluation (1,931)	54.63	72.61	68.98%

Table 3: Official leaderboard results by evaluation phase

Overall, our system reaches a chrF above 72 and a BLEU above 54 in both phases, together with a sentiment-preservation score of approximately 69% on the final official test leaderboard, indicating that the model reliably rewrites Arabic dialectal sentences into the opposite sentiment while retaining a substantial share of the original content.

5.1 Error Analysis

We manually inspected model outputs on the Development-phase validation set and identified two recurring patterns.

Exact or near-exact match with the human reference. For a large share of short, single-clause inputs, the model's output matches the gold target almost word-for-word. For example, the input "صادقه شباصه" (negative) is rewritten as "صادقه حلوة", identical to the gold target, showing the model has learned a reliable direct mapping for common short evaluative phrases. Similarly, for the input "الرساله لا تكفي لبوح كل مافي القلب" (negative), accompanied by a broken heart emoji, the gold target is "الرساله تكفي بكثير لبوح كل مافي القلب ♡", and our model generates "الرساله تكفي لبوح كل مافي القلب" with a sparkling heart emoji: the negation is correctly removed and the emoji is correctly flipped to a positive one, closely matching the intended meaning of the gold target.

Residual, unflipped cues leaving a mixed-signal output. A second pattern is that the model sometimes flips the main polarity-bearing word but leaves a secondary cue (an emoji) from the original polarity in place. For example, "يا محلي عمري والأيام ♡" (positive) is rewritten as "يا مضيع

عمري والأيام ♡”: the adjective is correctly inverted (sweetness → waster), but the heart symbol (♡) is carried over unchanged, whereas the gold target replaces it with a broken heart emoji. The resulting output therefore mixes a negative statement with a positive visual cue.

A likely contributor to the Sentiment Preservation drop between the Development and Final Evaluation phases (Table 3) is that the released Development-phase file included the gold `source_polarity` label for each example, while the Final Evaluation test file did not. For the final test submission we therefore had to predict `source_polarity` ourselves using the same CAMeL sentiment classifier before generation, so that the model would know which direction to invert. Any misclassification at this step directs the model to invert in the wrong direction, and this compounding error most likely explains a meaningful share of the gap between the two phases' Sentiment Preservation scores.

5.2 Limitations

Error propagation from predicted source polarity. As noted above, the Final Evaluation test file does not include gold `source_polarity`, so we predict it with an external sentiment classifier before generation. Misclassifications at this step propagate into the final output and likely explain part of the drop in Sentiment Preservation between the Development and Final Evaluation phases.

No data augmentation or candidate reranking. Our final system trains directly on the provided source→target pairs with standard beam search decoding (4 beams), without bidirectional data augmentation or multi-candidate reranking at inference time. It is possible that such techniques would further improve either content preservation or sentiment accuracy, but we did not adopt them in our submitted system.

Classifier-based sentiment scoring. Our reliance on a single external classifier (CAMeL-Lab's Arabic dialectal sentiment model) — both to infer `source_polarity` at test time and, per the task setup, for automatic sentiment scoring — means any systematic bias in that classifier affects our results and may not fully align with human judgment.

Secondary polarity cues. As shown in the Error Analysis, the model does not always propagate the sentiment flip to non-lexical cues such as emojis, which can leave a mixed-signal output even when the main wording is correctly inverted.

No human evaluation. All reported numbers are automatic (BLEU, chrF, classifier-based sentiment accuracy). We do not conduct human evaluation of fluency, dialectal naturalness, or perceived sentiment.

6 Conclusion

We presented NU-SentEval's system for DialectSentEval 2026 Subtask 2 (Arabic Dialect Sentiment Swap): a fine-tuned AraT5v2-base-1024 sequence-to-sequence model trained directly on the shared task's source-target pairs and decoded with standard beam search. Our system reaches 55.60 BLEU, 72.27 chrF, and 77.90% sentiment preservation in the Development phase, and 54.63 BLEU, 72.61 chrF, and 68.98% sentiment preservation on the official Final Evaluation test leaderboard. Our error analysis shows the model reliably inverts sentiment for short, single-clause inputs, often matching the human reference exactly, and frequently produces sentiment-correct rewrites that use different but equally valid wording than the gold target. The main remaining weaknesses are inconsistent handling of secondary polarity cues such as emojis, and error propagation from automatically predicted source polarity at test time, which we identify as the most likely contributor to the drop in sentiment-preservation score between the two evaluation phases.

7 Acknowledgments

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A Replication Details

We fine-tune UBC-NLP/AraT5v2-base-1024 for 8 epochs (batch size 16, learning rate 3e-4, weight decay 0.01, warmup ratio 0.06, fp16), using beam search decoding with 4 beams (max input/target length: 96 tokens). Training uses 6,752 examples (6,077 for gradient updates, 675 held out internally to monitor chrF for checkpoint selection), and completes in approximately 40 minutes. Sentiment labels (both for predicting `source_polarity` at test time and for automatic evaluation) are obtained from CAMeL-Lab/bert-base-arabic-camelbert-da-sentiment. Library versions: transformers 4.46.3, tokenizers 0.20.3, datasets 3.1.0, accelerate 1.1.1. The fine-tuned model is

publicly available at https://huggingface.co/areejkhaled33/arat5-sentiment-swap.

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